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# Amplifying Human Strengths Through AI

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# The Gap No One Is Talking About

From “We Have AI” to “AI Makes Us Better”

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## **Quick show of hands**

- How many of you have used AI at work this week?

## **The uncomfortable question**

- How many of you have team members using ChatGPT — and you don't know what they're feeding into it?

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## The real shift

- For decades, we could only automate structured work — spreadsheets, databases, form fields

## What LLMs changed

- Documents, emails, contracts, conversations — everything that used to need a human to read and interpret. We can now automate the messy stuff.



# The AI Reality Check

Where Most Organisations Actually Are  
With AI



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# How I did 1000 man hours work in 30mins

3 wheeler Auto  
Aadhar card  
awning extension  
backpack  
barricade  
Bicycle wheel  
Bucket  
chappals  
Cheetah  
Closed Book  
Cow  
Crocodile  
Crow  
Donkey  
Driving License  
Elevator  
Face  
Flower  
Fruit Shop  
Grill Gate  
Hair Dryer  
Ice cream parlour  
Iron box  
Jaguar (Animal)  
Key  
Laptop charger  
meat stall  
Mobile charger  
Mobile phone  
NameBoard-Bakery  
NameBoard-Beauty Parlour  
NameBoard-Books , toys , gift shop  
NameBoard-College  
NameBoard-courier office  
NameBoard-Departmental Store  
NameBoard-electronics home appliances

NameBoard-fancy store  
NameBoard-government offices  
NameBoard-jewellery shop  
NameBoard-library  
NameBoard-Mobile Shop  
Open Book  
Pan Card  
Parrot  
Passport  
Peacock  
Pedestal fan  
Pencil  
Petrol Bunk  
Pushcart  
Rolling chair  
Scooter  
seesaw  
Share Auto  
Sheep  
shoes  
Shop shutter  
shopping cart  
Sign board-Baby care/Feeding Room  
Sign board-Canteen/Cafeteria  
Sign board-Emergency exit  
Sign board-Entry  
Sign board-Escalator  
Sign board-Exit  
Sign board-Fire exit  
Sign board-Hotel/Lodge  
Sign board-Information desk  
Sign board-Lift (Elevator)  
Sign board-Medical shop/ Hospital/ First aid  
/ Medical aid  
Sign board-Police station  
Sign board-Reception  
Sign board-Restaurant  
Sign board-Restricted area  
Sign board-Stairs  
Sign board-Waiting area

SignBoard-Distance milestone  
SignBoard-Drinking water  
SignBoard-Emergency (Hospital sign inside  
buildings)  
SignBoard-Entry gate  
SignBoard-Exit gate  
SignBoard-Fire alarm  
SignBoard-Fire extinguisher  
SignBoard-Foot overbridge  
SignBoard-Fuel station  
SignBoard-Hospital ahead  
SignBoard-Men at work  
SignBoard-No entry  
SignBoard-No parking  
SignBoard-No smoking  
SignBoard-Parking  
SignBoard-Pedestrian crossing  
SignBoard-Railway crossing  
SignBoard-School ahead  
SignBoard-Speed breaker  
SignBoard-Stop  
SignBoard-Traffic signal ahead  
SignBoard-Washroom (Female)  
SignBoard-Washroom (Male)  
SignBoard-Zebra crossing  
slide  
Spectacles  
suitcase  
sweet stal  
swing  
Tea Stall  
Television  
Treadmills  
Turn Pages  
Vegetable shop  
Wallet/Purse  
Water filter  
White cane  
Wifi Router  
wrist watch

I am building an object detection model and need to find data sources for these classes with bounding boxes. Even if bounding boxes are not available, I can still use raw images and manually annotate them. Prioritize datasets that already include bounding boxes; if those are not available, raw image data is sufficient. Search across sources such as Kaggle, Hugging Face, GitHub repositories, Google Dataset Search, Roboflow datasets, and similar platforms. **1230 sources in 15 mins**

I'll launch into comprehensive research to find dataset sources for your 121 object classes. Let me search across multiple platforms to identify datasets with bounding boxes (prioritized) or raw images that you can annotate.

Dataset source research for object detection

Research complete • 1,230 sources • 15m 34s

Your comprehensive dataset research for 121 object detection classes is ready. It includes detailed coverage analysis across vehicles, documents, animals, electronics, signage, and Indian-specific items, with 75+ usable datasets identified containing over 500,000 annotated images. The report provides direct URLs, image counts, licensing information, bounding box availability, and highlights critical gaps requiring custom collection—particularly for water filters, Indian shop nameboards, and specialized signage.

Comprehensive Dataset Sources for 121 Object Detection Classes: Indian and...

Document

### Dataset source research for object detection

Research plan created >

Gathered 1,230 sources v

|                                                                                                             |             |             |
|-------------------------------------------------------------------------------------------------------------|-------------|-------------|
|  universe.roboflow.... | 247 sources | <div></div> |
|  kaggle.com            | 164 sources | <div></div> |
|  github.com            | 114 sources | <div></div> |
|  huggingface.co        | 93 sources  | <div></div> |
| ... 612 other sources                                                                                       |             |             |



# Summary Tables

## Coverage by Category

| Category                  | Total Classes | Excellent (1000+ imgs) | Good (100-1000) | Limited (<100) | None |
|---------------------------|---------------|------------------------|-----------------|----------------|------|
| Vehicles & Transportation | 6             | 3                      | 2               | 0              | 1    |
| Documents & Cards         | 4             | 0                      | 4               | 0              | 0    |
| Animals                   | 9             | 3                      | 3               | 2              | 1    |
| Personal Items            | 9             | 5                      | 2               | 1              | 1    |
| Electronics & Appliances  | 10            | 4                      | 4               | 2              | 1    |
| Books & Stationery        | 5             | 2                      | 1               | 0              | 2    |
| Furniture & Equipment     | 5             | 2                      | 1               | 0              | 2    |
| Shop Types & Buildings    | 18            | 0                      | 2               | 1              | 15   |
| Indoor Signboards         | 20            | 11                     | 0               | 0              | 9    |
| Outdoor Signboards        | 23            | 10                     | 6               | 0              | 7    |
| Miscellaneous             | 4             | 1                      | 1               | 1              | 1    |

## Indian-Specific Items Status

| Item Type                    | Dataset Availability    | Best Source                           | Notes                                  |
|------------------------------|-------------------------|---------------------------------------|----------------------------------------|
| 3-wheeler Auto               | ✅ Excellent             | IDD, IIIT Autorickshaw, Roboflow      | 15,000+ annotated images available     |
| Share Auto                   | ✅ Same as auto-rickshaw | IDD, Roboflow                         | Shared auto-rickshaw service           |
| Aadhar Card                  | ⚠️ Synthetic only       | Roboflow, Mendeley, GitHub generators | Privacy restrictions, 2,000+ synthetic |
| PAN Card                     | ⚠️ Limited              | Hugging Face, Roboflow                | 1,100+ images with restrictions        |
| Driving License              | ✅ Good                  | Roboflow (3,016 images)               | Indian-specific datasets available     |
| Chappals                     | ❌ None                  | Use sandal datasets                   | No dedicated chappals datasets         |
| Indian Birds (Crow, Peacock) | ⚠️ Raw images           | Indian Birds dataset                  | 1,500 per species, need annotation     |
| Indian Signboards            | ⚠️ Traffic only         | DataCluster Labs, IEEE                | Shop signboards not covered            |
| Tea Stall, Sweet Stall       | ❌ None                  | Custom collection needed              | Major gap                              |



# When AI Gets It Wrong

## Why Structure and Governance Matter

### 1. Project Overview

A production-grade web application that converts standard **2D product photography** (Front, Back, Left, Right views) into interactive, textured **3D furniture models** using the [REDACTED] Multiview AI engine. The tool will be integrated into [REDACTED]'s existing Shopify-based e-commerce ecosystem, enabling customers to rotate, zoom, and inspect furniture pieces in-browser before purchase.

### 2. Development Investment

| Component                               | Amount (₹)                    |
|-----------------------------------------|-------------------------------|
| Architecture, Backend & API Integration | ₹ 2,00,000                    |
| Frontend Development & 3D Viewer        | ₹ 1,50,000                    |
| Testing, QA & Deployment                | ₹ 1,00,000                    |
| Project Management & Documentation      | ₹ 50,000                      |
| <b>TOTAL DEVELOPMENT COST</b>           | <b>₹ 6,00,000 + GST (18%)</b> |

# When AI Gets It Wrong

## Why Structure and Governance Matter

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### A real story

- An **AI-generated deliverable** looked polished, professional, and completely wrong. The client caught it. The team hadn't verified a single claim.

### The pattern

- **AI output looks confident** even when it's wrong. The better the formatting, the less people question it.

### The punchline

- “If you don’t verify, you’re not using AI — **AI is using you.**”

### This isn't a technology problem

- It's a human judgment problem. And there's **research** that explains exactly why it happens.

# Thinking: Fast, Slow, **and** **Artificial**

The Tri-System Theory of Cognition (Shaw  
& Nave, Wharton 2026)



# How Your Brain Actually Makes Decisions

## Daniel Kahneman's Two Systems of Thinking

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### System 1 — Fast Thinking

- Your autopilot. **Instant, automatic, effortless.** Recognising faces, reading words, slamming brakes. Runs on gut feelings and shortcuts.

### System 2 — Slow Thinking

- Your deliberate mode. **Takes effort, concentration, and energy.** Solving  $37 \times 24$ , comparing insurance plans, planning a project timeline.

### The key insight

- We think we're rational (System 2). In reality, System 1 runs the show most of the time. **System 2 often just rubber-stamps what System 1 already decided.**



# A Quick Test for Your System 1

## The Bat and Ball Problem

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### The question

- A bat and ball cost ₹110 together. The bat costs ₹100 more than the ball. How much does the ball cost?

# A Quick Test for Your System 1

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### The question

- A bat and ball cost ₹110 together. The bat costs ₹100 more than the ball. How much does the ball cost?

### Your System 1 screams: ₹10!

- But engage System 2 and do the math — the answer is ₹5. (Ball = ₹5, Bat = ₹105, Total = ₹110)

### Why this matters

- Knowing when your fast brain is helping vs fooling you — so you know when to slow down. Now imagine adding AI as a third system...

# Tri-System Theory of Cognition

How AI Reshaping Human Reasoning



# Tri-System Theory of Cognition

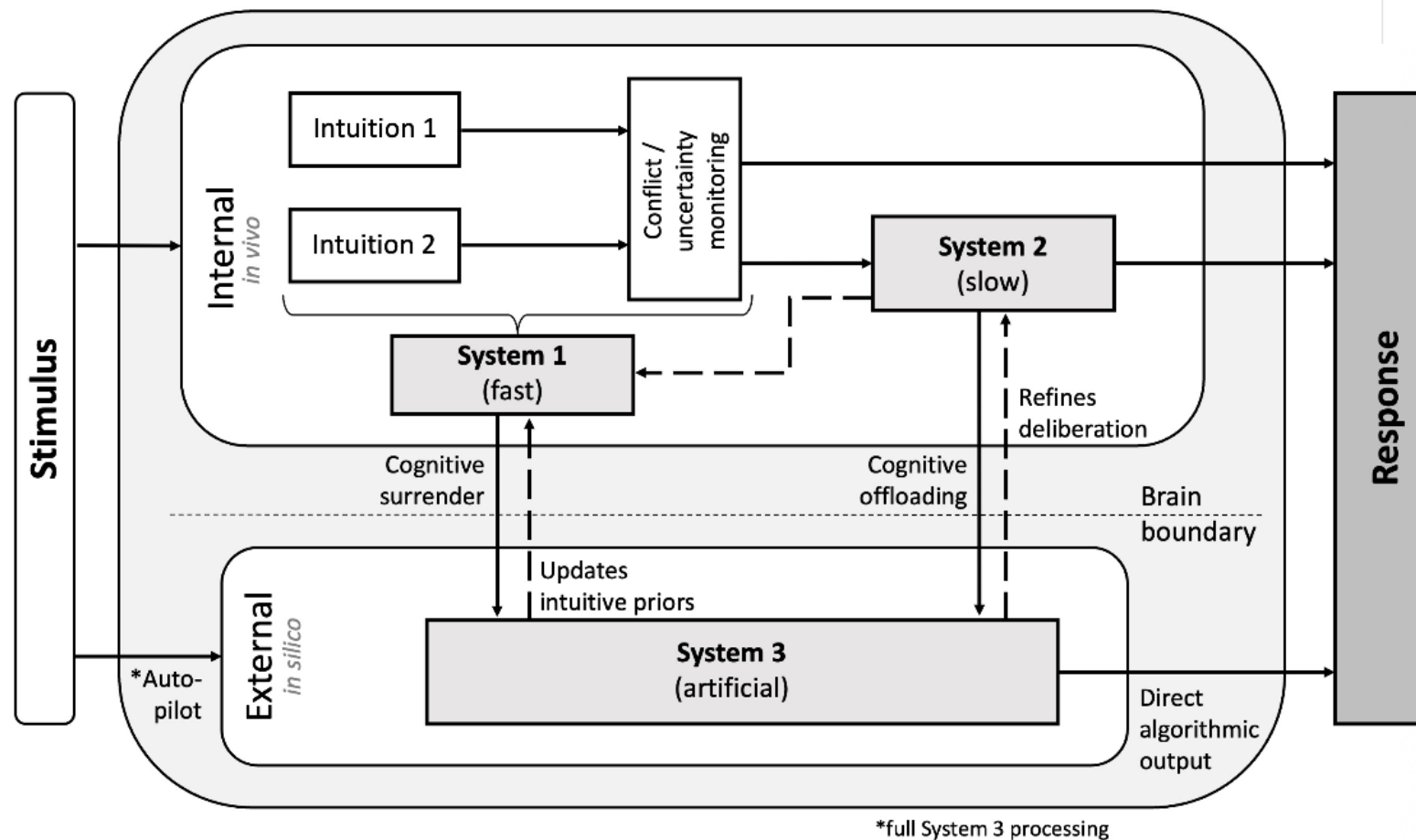
Source: Shaw & Nave, Wharton School, 2026

Thinking—Fast, Slow, and Artificial: How AI is Reshaping Human Reasoning and the Rise of Cognitive Surrender

[https://doi.org/10.31234/osf.io/yk25n\\_v1](https://doi.org/10.31234/osf.io/yk25n_v1)

*The Wharton School Research Paper*

58 Pages • Posted: 2 Feb 2026 • Last revised: 10 Feb 2026





# The Three Systems of Thinking

Source: Shaw & Nave, Wharton School, 2026

| Attribute         | System 1 (Fast)          | System 2 (Slow)           | System 3 (Artificial)             |
|-------------------|--------------------------|---------------------------|-----------------------------------|
| Origin            | Human (intuitive)        | Human (analytical)        | Artificial (algorithmic)          |
| Processing speed  | Fast                     | Slow                      | Fast/Variable                     |
| Cognitive effort  | Low                      | High                      | None/Variable                     |
| Accuracy          | Prone to bias            | Normative but effortful   | High in structured domains        |
| Affective input   | Emotion-driven           | Emotion-regulated         | Emotion-neutral                   |
| Ethical reasoning | Implicit norms           | Explicit deliberation     | Depends on training data          |
| Justification     | Experiential or post hoc | Rationalized, articulated | Data-driven, externally generated |

# What System 3 Brings to the Table

## The Cognitive Affordances — and Costs — of AI

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### Fast, accurate, and emotion-free

- System 3 delivers **data-driven, automated reasoning with minimal effort**. Neutral, nonpartisan outputs. In **fact retrieval, translation, and pattern recognition, it outperforms both System 1 and System 2**.

### But it comes at a cost

- **System 3 lacks affect, situational judgment, and normative reasoning grounded in human experience**. It simulates coherence from data rather than possessing true understanding.

### Not just a tool — a co-agent in reasoning

- System 3 is a qualitatively different mode of cognition: **externalised, automated, data-driven**. Humans access it on demand. It delivers outputs with epistemic authority.

# Cognitive Surrender

The Risk No One Is Talking About



# Cognitive Surrender in Numbers

1,372 participants | 3 studies | 9,593 trials

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## **80% followed faulty AI recommendations**

- When AI gave a wrong answer, 4 out of 5 people adopted it as their own

## **Accuracy swung by 40 percentage points**

- **Brain-only: 46% | AI-Accurate: 71% | AI-Faulty: 31%**

## **Confidence increased by 11.7 points even when AI was wrong**

- People felt more certain while being more wrong

## **Time pressure + faulty AI = 12.1% accuracy**

- Deadline pressure combined with unchecked AI is dangerous for project teams

## **Incentives + feedback doubled override rates (20% to 42%)**

- But the surrender gap persisted at ~44 percentage points

# Who Surrenders More?

Individual differences from the research

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## Higher Trust in AI

- More chat use and lower accuracy
- 3.5x more likely to follow faulty AI (OR = 3.47)
- Most vulnerable group in your teams

## Higher Fluid IQ

- Better accuracy on faulty trials
- Less likely to follow faulty AI (OR = 0.23)
- Natural resistance to surrender

## Higher Need for Cognition

- Better accuracy on faulty trials
- Reduced AI chat usage
- People who enjoy thinking resist surrender



# Cognitive Offloading vs Cognitive Surrender

The goal: move your teams from surrender to offloading

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## Cognitive Offloading

- Strategic delegation to AI
- System 2 stays engaged
- User checks and verifies AI output
- AI extends your thinking

## Cognitive Surrender

- Uncritical abdication to AI
- System 2 is bypassed
- User accepts without checking
- AI replaces your thinking

# Structured AI **as** **an Antidote**

From Cognitive Surrender to Cognitive  
Offloading



# GStack: Role-Based AI Workflows

Created by Garry Tan (YC CEO) | 20,000+ GitHub stars | MIT License

## Created by Garry Tan (YC CEO)

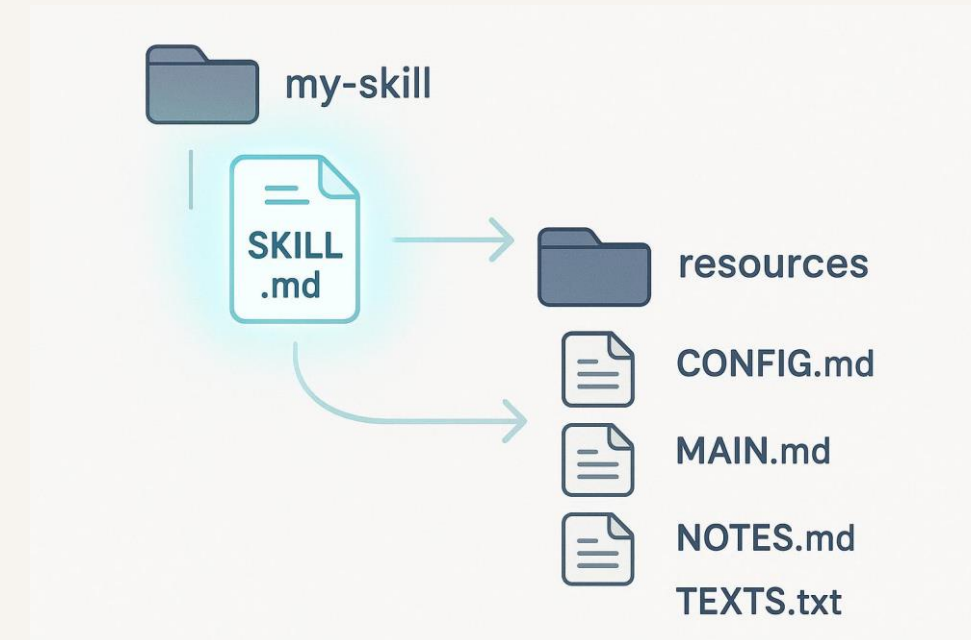
- Open-sourced March 2026, 20,000+ GitHub stars, MIT license
- A role-based AI workflow framework that turns AI into a structured engineering team

## Role-based AI for structured workflows

- CEO thinking, Eng review, Ship, QA — each role reviews from a different angle

## The SKILL.md pattern

- Encode your team's standards into AI guardrails — cognitive offloading by design
- This is cognitive offloading by design, not surrender by default



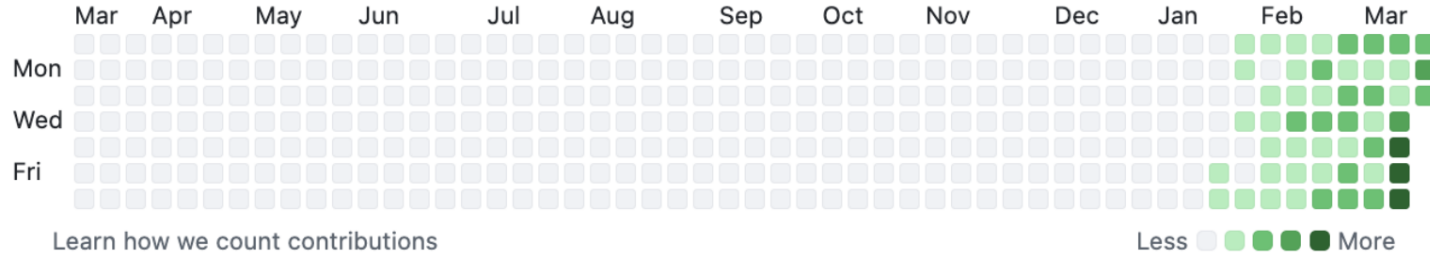
"I don't think I've typed like a line of code probably since December, basically, which is an extremely large change." — [Andrej Karpathy](#), No Priors podcast, March 2026

When I heard Karpathy say this, I wanted to find out how. How does one person ship like a team of twenty? Peter Steinberger built [OpenClaw](#) — 247K GitHub stars — essentially solo with AI agents. The revolution is here. A single builder with the right tooling can move faster than a traditional team.

## 2026 — 1,237 contributions and counting:

### 1,237 contributions in the last year

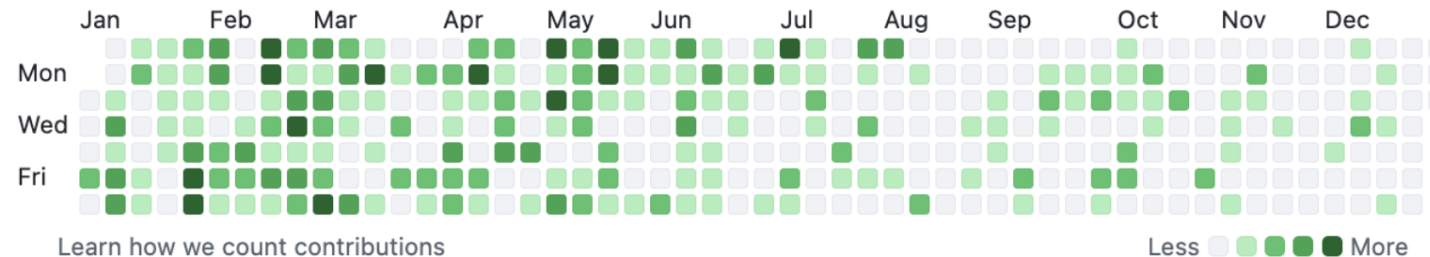
Contribution settings ▾



## 2013 — when I built Bookface at YC (772 contributions):

### 772 contributions in 2013

Contribution settings ▾



# Vibe Coding → To Viable Coding

## Spec Driven Development

# Spec-Driven Development

From specifications to implementation—powered by AI agents and human oversight

GitHub Spec Kit

BMAD-METHOD

AI-Assisted Coding

Agentic Workflows



## SPECIFY

Define requirements, user stories, acceptance criteria



## PLAN

Architectural design, database design, & tech stack decisions



## TASKS

Break into granular, actionable work items



## IMPLEMENT

AI-generated code with human review & refinement



# From Chatbots to Autonomous AI

The 5-Level Autonomy Spectrum



# The 5-Level AI Autonomy Spectrum

From ad hoc chat to autonomous agents — where the industry is in 2026

| Level             | What It Is                                                | Examples                      | Your Role    |
|-------------------|-----------------------------------------------------------|-------------------------------|--------------|
| L1: Chatbots      | You type, it responds. No memory, no actions.             | ChatGPT, Gemini, base Claude  | Operator     |
| L2: Copilots      | AI sees your context, suggests inline. You drive.         | GitHub Copilot, Cursor        | Collaborator |
| L3: Task Agents   | Assign a task, approve the result. AI plans and executes. | Devin, Claude Code            | Approver     |
| L4: Computer Use  | AI sees your screen, clicks buttons, navigates apps.      | Claude Computer Use, Operator | Supervisor   |
| L5: Autonomous AI | AI runs 24/7, acts proactively, spawns sub-agents.        | OpenClaw                      | Observer     |

# OpenClaw: When AI Doesn't Wait for You

## The Shift from Session-Based to Always-On AI

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### What is it?

- Open-source autonomous AI agent by Peter Steinberger. 310,000+ GitHub stars. Runs as a persistent daemon — 24/7.

### Jensen Huang's endorsement

- Nvidia CEO called it “the next ChatGPT.”

### How it works

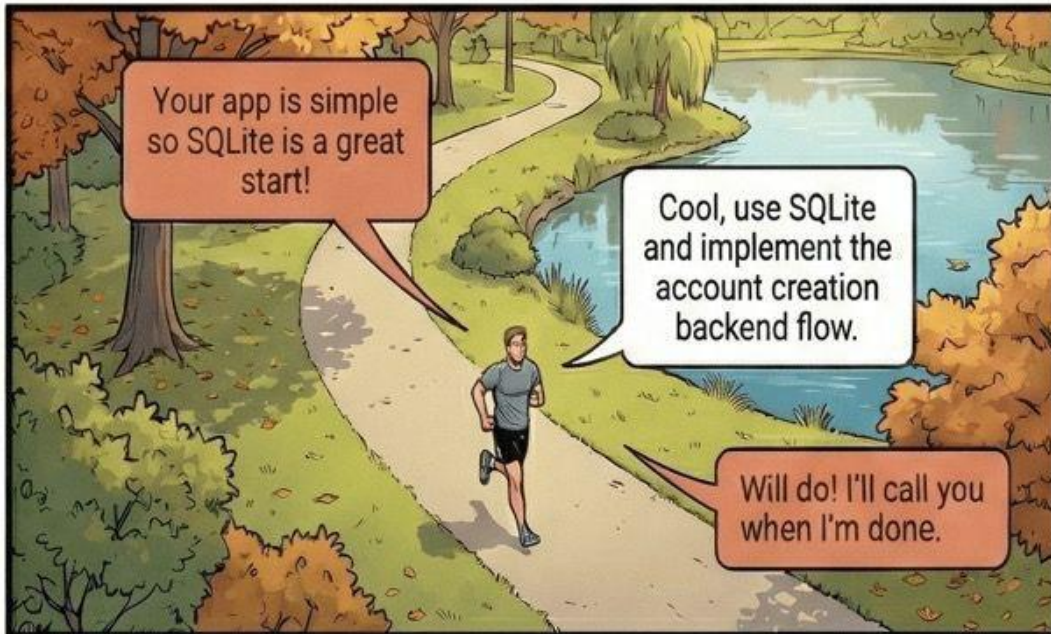
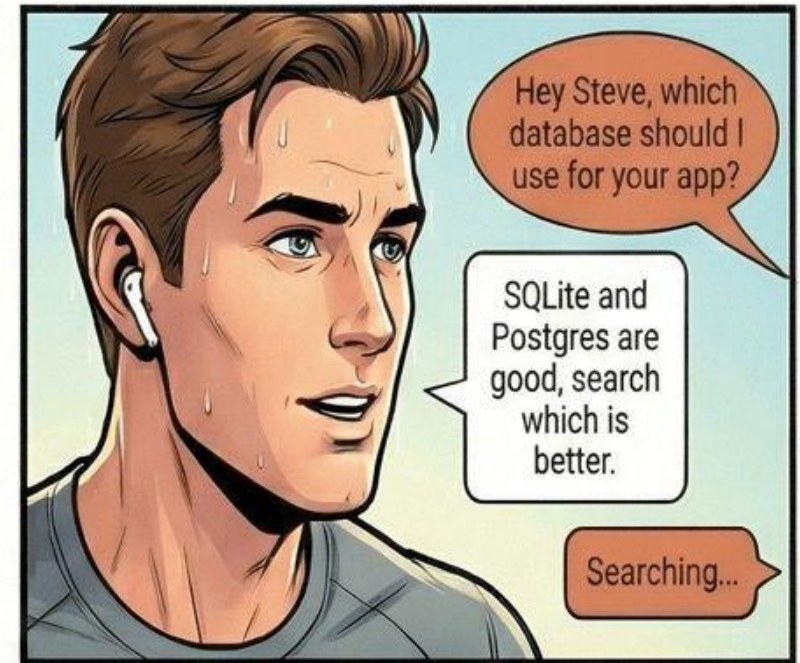
- Connected to 12+ messaging platforms. Browser control, cron jobs, memory that persists. AI that acts without being asked.

### What this means

- Imagine AI agents that monitor dashboards 24/7, draft status reports, and route messages — all without human prompting.









# AI for Everyday Project Delivery

Practical Tools Your Team Can Use  
Tomorrow





# AI for Everyday Project Delivery

Practical use cases you can try today

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## **Project Planning**

- Generate WBS, estimate effort, identify dependencies with structured prompts

## **Risk Assessment**

- AI analyzes risk register + timeline. PM reviews and validates. 10 min vs 45 min.

## **Status Reporting**

- Summarise data from project tools into executive updates automatically

## **Stakeholder Communications**

- Draft emails, meeting agendas, escalation notes with project context

## **Meeting Summarisation**

- Convert transcripts to action items. Never miss a follow-up.

# Where Are You on the AI Maturity Curve?

## A Self-Assessment for Your Organisation

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### **Level 1: Ad Hoc**

- “We use ChatGPT sometimes.” No structure, no consistency, no accountability.

### **Level 2: Structured**

- AI workflows exist. Role-based prompting (GStack pattern). Team standards encoded.

### **Level 3: Governed**

- Feedback loops in place. AI outputs verified. Accuracy tracked. ISO 42001 principles applied.

### **Level 4: Amplified**

- AI extends your team's thinking, not replaces it. Cognitive offloading by design.

# AI Governance

Keeping System 2 Alive in Your  
Organisation



# ISO/IEC 42001:2023 and AI Design Principles

Nunnari Labs is among the first companies in India to be certified

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## ISO 42001: AI Management System

- How your organisation identifies and manages AI risks
- How you ensure AI systems are fair, transparent, and accountable
- Think of it as ISO 9001 but for AI

## Design Recommendations from the Research

- Adaptive nudging: AI interfaces that signal uncertainty
- Confidence indicators: showing when AI is uncertain vs confident
- Customisable autonomy modes: letting users choose AI involvement
- Digital/AI literacy: equipping teams to know when to trust AI



# Three Questions Every Project Leader Should Ask

A practical framework for your organisation

1

Step 01

Where is AI?

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Map every point where AI touches your team's decisions

2

Step 02

Offloading or Surrender?

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Is your team using AI as a tool or accepting outputs uncritically?

3

Step 03

What's Your Feedback Loop?

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What mechanisms does your team have to catch AI errors?



# Key Takeaways

From Cognitive Surrender to Cognitive Amplification

LLMs unlocked automation on unstructured work — that changed everything

Cognitive surrender is real — 80% follow faulty AI without questioning

Structure beats surrender — role-based AI keeps judgment engaged

AI autonomy is accelerating — governance is no longer optional

Start now — map, structure, verify. One workflow at a time.

# Nunnari Labs & AI Tamil Nadu

AI, with Soul.

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## **Nunnari Labs (Founded 2020, Coimbatore)**

- Enterprise AI, Edge AI, Computer Vision, Assistive Technology
- ISO/IEC 42001:2023 certified — among the first in India
- 20+ core engineering team across AI/ML, full-stack, and research

## **AI Tamil Nadu Community (Founded 2018)**

- 10,000+ members across Tamil Nadu
- 8+ years of meetups, workshops, and open-source projects

## **Nunnari Academy & Research**

- AI education in Tamil + English | Edge AI + Indic language LLM/TTS/STT

## **Connect With Us**

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Thank You!  
Q&A?

